

TESFA WORKU MESHESHA, PhD

Water Resources Scientist | Hydrology | Watershed Science | GIS & Remote Sensing

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PROFESSIONAL SUMMARY

Hydrologist and Research Scientist with a Ph.D. in Hydrology and experiences in water resources modeling, natural resources management, groundwater modelling, watershed analysis, and environmental systems research. Expertise in hydrologic and water quality modeling, drought and climate impact assessment, and GIS/remote sensing for watershed and source water studies. Proven ability to develop decision-support tools integrating large-scale environmental datasets, hydrologic simulation, and statistical/ML approaches to support water resources planning, ecosystem protection, and climate resilience.

CORE SKILLS

- **GIS & Remote Sensing:** ArcGIS Pro, QGIS, Google Earth Engine, spatial analysis, watershed delineation, environmental mapping.
 - **Modeling Tools:** SWAT, MODFLOW, HEC-RAS, statistical hydrology, uncertainty analysis, machine learning for environmental systems.
 - **Water Resources & Hydrology:** Watershed modeling, hydrologic simulation, water quality modeling, drought analysis, reservoir and streamflow assessment, climate change impact analysis
 - **Programming & Data Science:** Python, R, MATLAB, FORTRAN, data analytics, AI/ML applications
 - **Data & Database Systems:** PostgreSQL, MySQL; statistical analysis & uncertainty quantification; machine learning for environmental systems
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PROFESSIONAL EXPERIENCE

Assistant Research Scientist

NASA Goddard Space Flight Center | Aug 2024 – Present

- Developed terrestrial ecosystem and hydrologic models to quantify carbon and nutrient fluxes using large-scale satellite and in-situ datasets.
 - Built data-driven hydrologic and ecosystem modeling frameworks integrating machine learning for drought monitoring, water resources forecasting, and watershed decision support.
 - Developed near-real-time environmental prediction systems for hydrologic and carbon cycle monitoring to support climate and water resource assessments.
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Assistant Project Scientist

- Modeled wildfire impacts on hydrologic processes including runoff, sediment transport, and post-fire water quality degradation.
 - Developed predictive models using machine learning and process-based hydrology for climate-resilient watershed and water system analysis.
 - Produced decision-support tools for evaluating water availability and ecosystem response under climate stressors.
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Postdoctoral Associate

City University of New York & NYC Department of Environmental Protection | Jan 2021 – Feb 2023

- Developed watershed management and water quality models under climate variability, focusing on erosion, nutrient transport, and hydrologic response.
- Applied process-based and statistical models to evaluate pollutant sources, transport mechanisms, and watershed-scale environmental impacts.
- Supported development of sustainable watershed management strategies for urban water supply systems.

Postdoctoral Researcher

Athabasca University | Mar 2019 – Nov 2020

- Enhanced SWAT hydrologic model for improved simulation of streamflow, sediment transport, and water quality under climate change scenarios.
- Developed long-term climate and hydrologic datasets to support water resources planning and watershed-scale modeling.
- Quantified ecosystem responses including nutrient loads, greenhouse gas emissions, and hydrologic variability.

SELECTED PUBLICATIONS

- Lopez et al., 2025. Post-wildfire sediment fluxes and turbidity dynamics in coastal watersheds. *Earth and Space Science*.
- Meshesha et al., 2025. Advancing global river discharge estimation using integrated remote sensing and modeling. AGU Meeting.
- Meshesha et al., 2021. Groundwater quality modeling in Athabasca River Basin using modified SWAT. *Scientific Reports*.
- Melaku et al., 2022. Carbon dioxide and ecosystem exchange modeling using SWAT. *Water*.

(Full list of publications: [Google Scholar](#))

EDUCATION

- Ph.D., Hydrology & Watershed Management
- M.Sc., Water Resources Engineering and Management
- B.Sc., Geospatial Science

CERTIFICATIONS

Python for Data Science | R Programming | SQL & Databases | Big Data in Water Resources | AI Applications in Environmental Systems

REFERENCES

1. Dr. Cecile Severine Rousseaux, Senior Scientist, Goddard Space Flight Center, NASA
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2. Dr. Erin Hestir, Professor, University of California, Merced; Phone: 510 362 6347
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